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New Mesozoic paleomagnetic results from the northeastern Sichuan basin and their implication

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ABSTRACT

This paper presents the results of a paleomagnetic study undertaken on samples from Lower Triassic and Lower Cretaceous sediments within the northeastern Sichuan Basin, China, in an area close to the Daba Mountains. Stepwise thermal demagnetization was used to isolate the characteristic higher temperature component (HTC) for these samples. These HTC (T1: D = 53.7°, I = 21.6°, α_{95} = 6.5°; K1: D = 28.9°, I = 25.7°, α_{95} = 6.6°) passed both fold and reversal tests, and indicate that these sediments have undergone sequential clockwise rotations relative to the Sichuan basin. These detected tectonic rotations were probably caused by the obstruction of the Hannan massif during Mesozoic thrusting within the southern Daba Mountains. The northeastern Sichuan Basin and the central Yangtze Fold Belt formed part of a continuous belt of basins (the northern Yangtze Basin), and acted as a coherent tectonic domain during the Early–Middle Jurassic. The Daba Mountains and the central Yangtze Fold Belt subsequently underwent rapid uplift and cooling around the Late Jurassic and Early Cretaceous. This suggests that tectonic rotations within both the central Yangtze Fold Belt and the Daba Mountains were very probably caused by intra-continental deformation over a long period of time rather than a single tectonic event.

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1. Introduction

The collision between the North China Block (NCB) and South China Block (SCB) is one of the most important geological events during the Mesozoic formation of East Asia. This process is thought to have finished by the Early–Middle Jurassic (Yang et al., 1992; Yang and Besse, 2001), and subsequent compressional deformation of the Yangtze Block during the Late Jurassic and Cretaceous was associated with the development of a Mesozoic thrust system that extended from the Xuefeng and Wuling mountains in the southeast, to the Sichuan Basin in the northwest (Yan et al., 2003). This area is far from any present-day continental margins, and the majority of the post-Late Triassic deformation occurred in an intracontinental setting (Liu et al., 2012). These observations suggest that deformation in the Yangtze block after the NCB–SCB collision was not limited to the margins.

Recent studies of the Middle Yangtze Fold Belt have identified clear clockwise rotations, as indentified using paleomagnetic data obtained from Triassic strata (Huang and Opdyke, 1996, 1997; Su et al., 2005; Tan et al., 2007). However, the processes that caused this rotation are unclear. For example, Huang and Opdyke (1996, 1997) determined that the Middle Triassic paleomagnetic declination from the central Yangtze Fold Belt is roughly paralleled to E–W trending fold axes, although samples taken closer to the Qinling suture yielded northerly paleomagnetic declinations. Thus, Huang and Opdyke (1997) suggested that these rotations were not related to the NCB–SCB collision. In addition, Yang and Besse (2001) tested the oroclinal–rotation hypothesis within the Yangtze Fold Belt by comparing paleomagnetic declinations and strike directions for the Mesozoic fold axes in latest Permian to Middle Triassic sediments, concluding that this hypothesis was negative. Furthermore, Tan et al. (2007) compared paleomagnetic rotations with fold axis trends within the Yangtze Fold Belt and suggested that the general clockwise rotation was in fact related to the NCB–SCB collision, although the timing of the clockwise rotation is not well constrained, and further studies are needed to determine whether the central Yangtze Fold Belt is in fact an orocline. This question cannot be answered until data are obtained from younger sediments in this area. However, analysis of Jurassic sediments within the Zigui area did not yield primary paleomagnetic directions (Tan et al., 2007) Therefore, we tried to find new evidences from other areas near the northwestern margin of the SCB.

The Daba Mountains form a curved tectonic belt within the southern Qinling Orogen; the southward bend in these mountains contrasts sharply with the rest of this orogenic. In the Daba fold belt, the Chengkou–Zhongbao fault has been a key boundary between the Northern and Southern Daba Mountains (Fig. 1). Many geological studies were carried out in this area (Dong et al., 2006; S. Dong et al., 2008; Y. Dong et al., 2008; Guo et al., 1996; He et al., 1997, 1999; Hu et al., 2012; Liu et al.,

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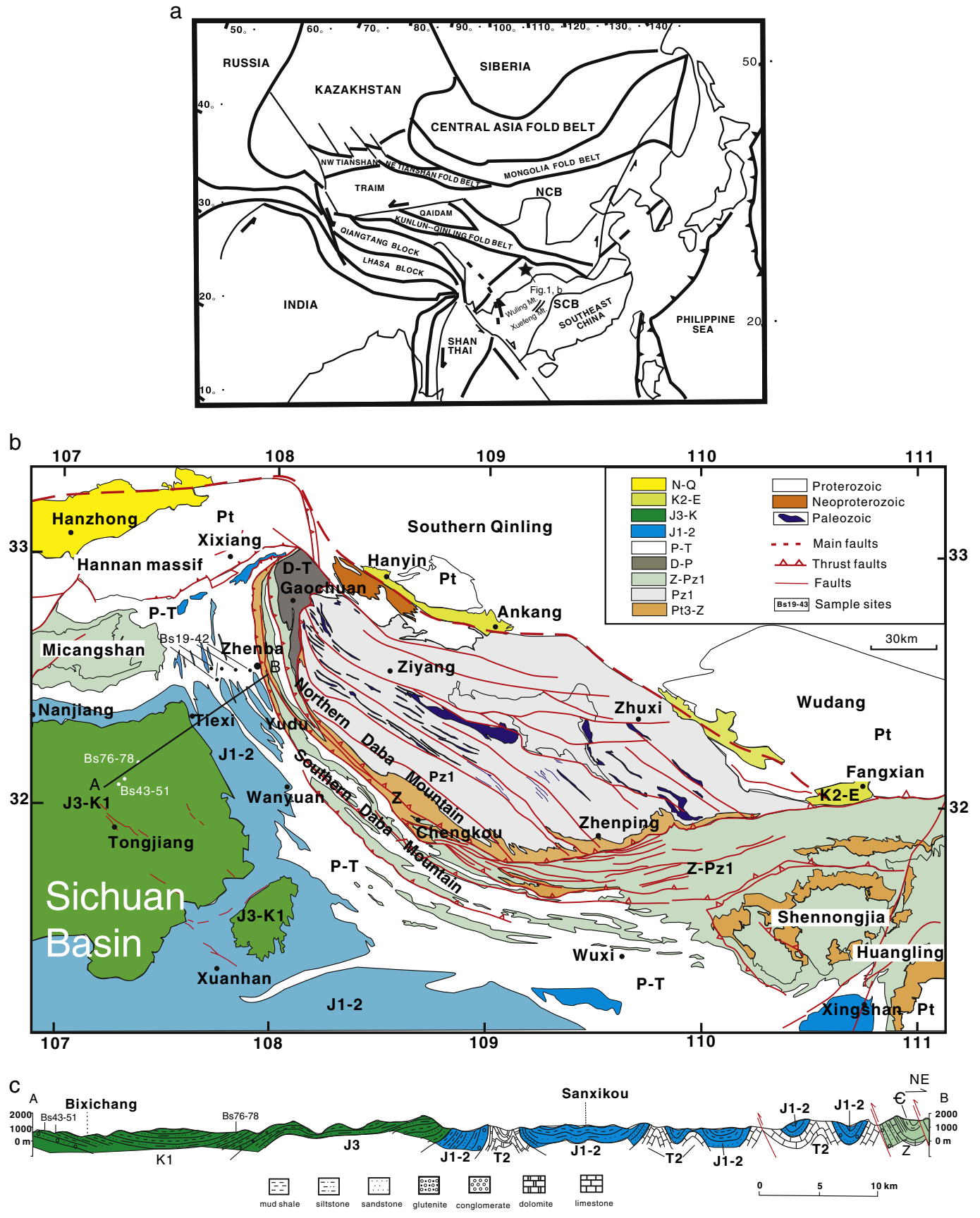


Fig. 1. (a) Simplified tectonic map of China and adjacent area (modified from Yang et al., 1992). The star denotes the sampling area. (b) Simplified geological map of the Daba Mountain belt and its neighboring areas (modified from Zhang et al., 2010). (c) Cross-sections of the study area (modified from unpublished 1:200,000 geological map).

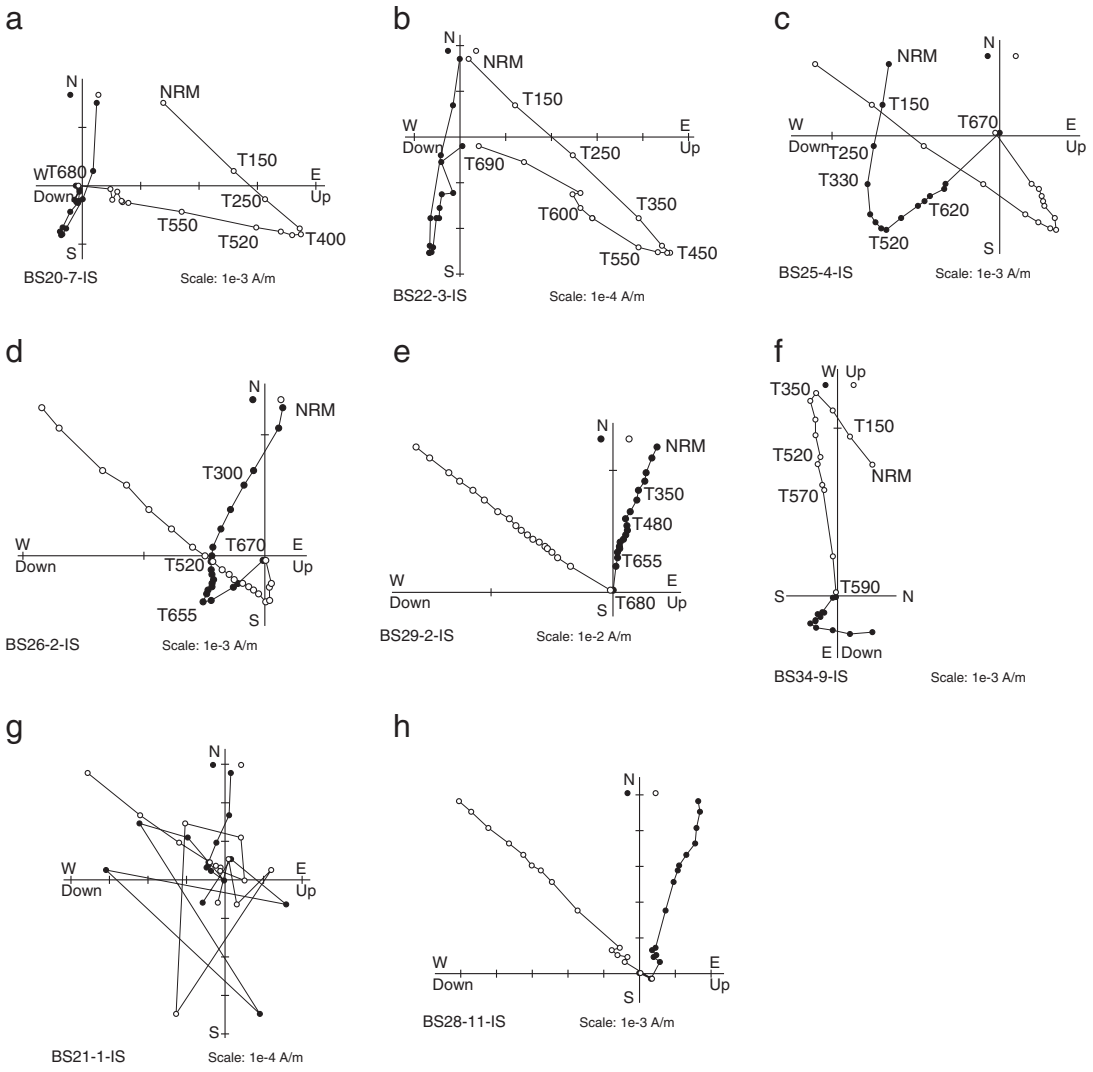


Fig. 2. Representative Zijderveld diagrams of thermal demagnetization of natural remanent magnetism (NRM) from the Triassic samples (in in situ coordinates).

2006; Shen et al., 2007; Wang et al., 2004, 2006; Wang and Zhang, 2010; Yue, 1988; Zhang et al., 2009; Zhang, et al., 2010). It is generally accepted that intense Late Jurassic–Early Cretaceous intracontinental tectonism occurred within the southern Qinling Orogen and the Northern Daba Mountains. This deformations affected Mesozoic sedimentary cover sequences in this area, and thrusting and folding occurred throughout almost the entirety of the southern Daba Mountains. However, this southwestward-directed thrusting was obstructed by the rigid Hannan and Huangling–Shennongjia massifs to the west and east, respectively, resulting in the formation of an arcuate structure in this area (Guoet al., 1996; Liu et al., 2006; Zhang et al., 2010).

Previous study of the Early–Middle Jurassic sediments in the Southern Daba Mountains (Wang and Zhang, 2010) provided evidence of a Late Jurassic–Early Cretaceous clockwise rotation (26.5 ± 9°). The curved tectonic belts within the Daba Mountains and the central Yangtze Fold Belt have a number of similarities, including: (1) a close relationship to the NCB–SCB collision, (2) evidence of late Mesozoic tectonic rotations, (3) location within the margins of the Yangtze block, and (4) both areas contain Late Triassic and Early Jurassic sediments. Given these similarities, here we present a new model for tectonic rotations within the central Yangtze Fold Belt. This model is based on data obtained from Mesozoic sediments near the Daba Mountains, and provides insights into the tectonic evolution of the Qinling Orogen and the post-collisional processes that affected both the North and South China plates.

2. Geological setting and sampling

Paleomagnetic samples were obtained using a portable drill and were collected from the Lower Triassic Feixianguan and Lower Cretaceous Cangxi formations along roads between Zhenba County in Shaanxi Province and Tongjiang city in Sichuan province. The Feixianguan Formation consists of aubergine-colored marlite that contains Early Triassic *Claraia aurita*, *Claraia stachei*, and *Eumorphotis mulliformis* fossils, whereas the Cangxi Formation consists of brick red and aubergine-colored mudstones and siltstones (BGMRS, 1991).

A total of 24 sites were drilled (13 sites within the Feixianguan formation and 11 sites within the Cangxi formation), all of which were oriented with a magnetic compass (Fig. 1). Samples were cut into 2.2-cm-long specimens at the Paleomagnetic Laboratory of Nanjing University, China, and present-day geomagnetic declinations were determined (−2.6°at Zhenba County) using the International Geomagnetic Reference Field (Mandea and MacMillan, 2000).

3. Paleomagnetic results

All measurements were carried out at the Paleomagnetic Laboratory of Nanjing University. Thermal-magnetic process was undertaken using a KLY-3s Kappabridge. Stepwise thermal demagnetization up to 690 °C was undertaken using an ASC TD48 oven and natural remanent magnetization (NRM) was measured using a 2G-755 cryogenic magnetometer

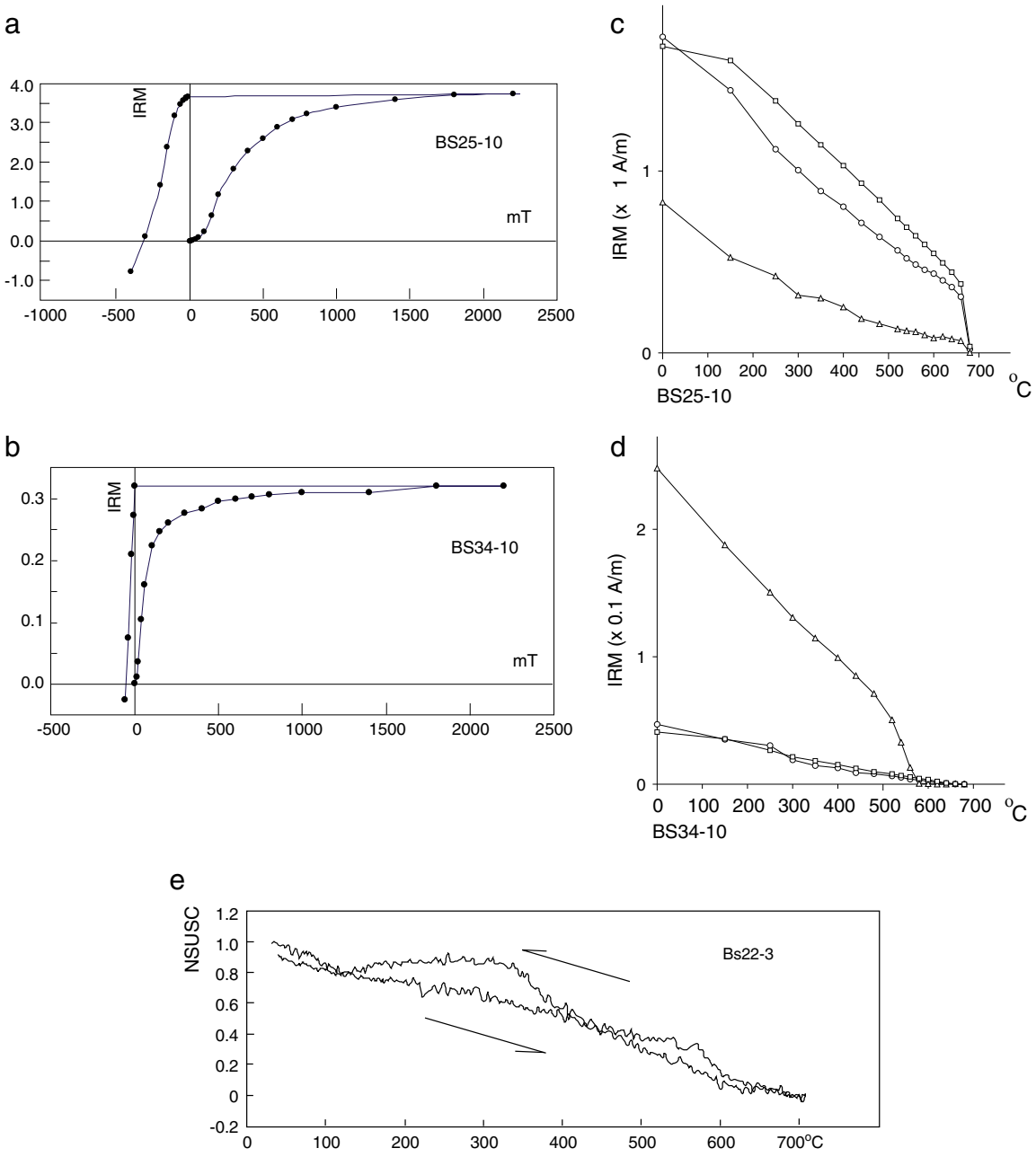


Fig. 3. (a) and (b) Isothermal remanent magnetization (IRM) acquisition curves of the Triassic samples. (c) and (d) Three component IRM thermal demagnetization showing unblocking temperature around 680 and 580 °C, respectively. (e) Temperature dependence of magnetic susceptibility when heating from room temperature to 600 or 700 °C and subsequently cooling process. The results of thermal-magnetic process indicated that few distinct chemical changes occurred during heating.

installed in a shielded room. Magnetic directions were analyzed using principal component analysis (Kirschvink, 1980), and the site-mean direction was calculated using Fisherian statistics (Fisher, 1953). Sister samples were selected for the acquisition of isothermal remanent magnetization (IRM) and demagnetization of composite IRMs in order to identify magnetic carriers. The demagnetization used three different fields that were applied along three orthogonal axes (0.12 T, 0.4 T and 2.2 T along the X, Y and Z axes, respectively) using the experimental procedures described by Lowrie (1990).

3.1. Early Triassic samples

A total of 146 samples were collected from the Feixianguan formation, with 63 samples yielding two steady components after thermal demagnetization (Fig. 2). A total of 43 of these samples had higher unblocking temperatures (680 °C–690 °C) whereas the other 20 samples had showed lower unblocking temperatures (580 °C–590 °C). Rock magnetic tests indicate that hematite and magnetite are the dominant carriers for these two groups of samples, respectively (Fig. 3). Samples with higher unblocking temperatures have IRM acquisition curves that rapidly increase below 1000 mT, but do not reach saturation even at 2.2 T; in comparison samples with lower unblocking temperatures reached saturation at about 1000 mT.

Low temperature components (LTC) were separated below 300 °C–655 °C and 300 °C–400 °C classes associated with hematite and magnetite magnetic carriers, respectively (Fig. 2). These components have a site-mean directions (before tilt correction: Dg/Ig = 11.4/50.8°, kg = 105.4, α_{95} = 4.2°; after tilt correction: Ds/Is = 46.1/23.7°, ks = 24.4, α_{95} = 9.0°) (Fig. 4a, b) that failed the fold test and are closed to present geomagnetic field, suggesting that the LTC may represent viscous

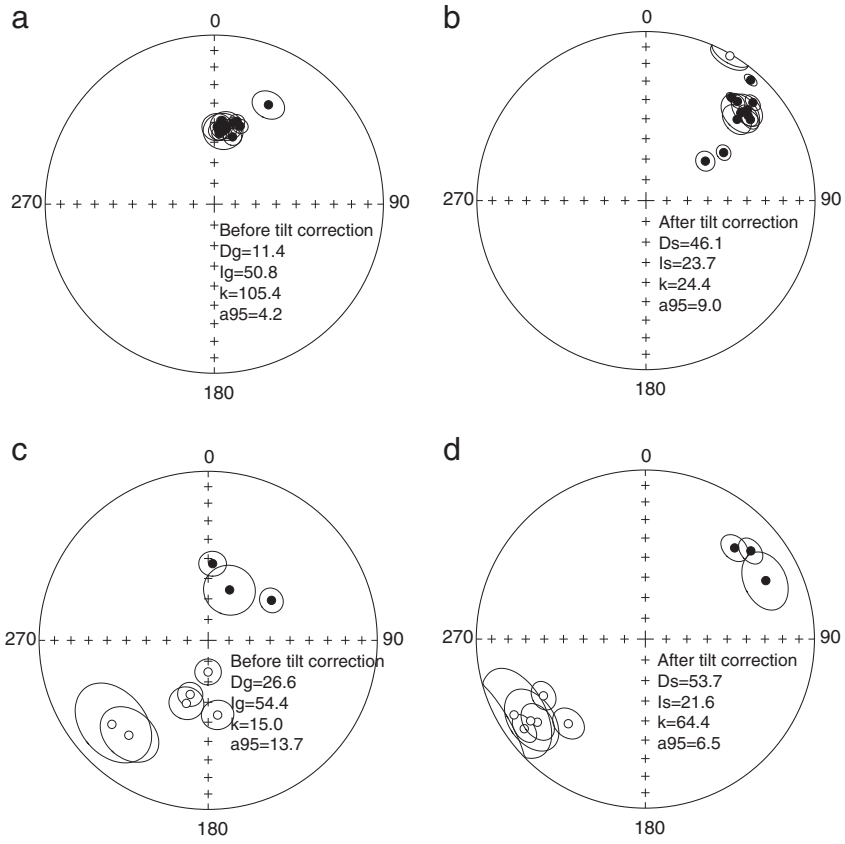


Fig. 4. Equal-area stereographic projections of low temperature components (LTC) (a, b) and high temperature components (HTC); (c, d) site-mean directions from the Triassic samples before and after tilt correction.

remanent magnetization. High temperature components (HTC) were separated into 680 °C–690 °C and 580 °C–590 °C classes according to unblocking temperatures. These components have site-mean directions (before tilt correction: Dg/Ig = 26.6/54.4°, kg = 15.0, α_{95} = 13.7°; after tilt correction: Ds/Is = 53.7/21.6°, ks = 64.4, α_{95} = 6.5°) (Fig. 4c, d) (Table 1) that passed both fold and reversed tests (Enkin, 2003; McFadden, 1990; McFadden and McElhinny, 1990), with the maximum precision parameter of the mean HTC direction occurring at 100% unfolding. This indicates that the Lower Triassic HTC direction was the characteristic remanent direction acquired during sedimentation.

3.2. Cretaceous samples

Rock magnetic experiments indicate that hematite was the dominant carrier of magnetic mineral in the red samples from the lower Cretaceous sediments. Their IRM acquisition curves did not reach saturation even at 2.2 T. Subsequent thermal demagnetization yielded a high unblocking temperature of 680 °C–690 °C, and thermal-magnetic analysis indicated that few chemical changes occurred during heating (Fig. 5).

A few Cretaceous samples showed erratic directions or directions that did not trend towards the origin during thermal demagnetization,

Table 1
Paleomagnetic results of characteristic directions from early Triassic strata in the Zhenba area.

Site	Slat.	Slong.	Strike/dip	N/n	Dg	Ig	Ds	Is	*k*	α_{95}
Bs20	32°31′54″	107°40′57″	347/62	7/14	198.4	−62.4	233.5	−11.9	104.4	5.9
Bs22	32°31′29″	107°41′8″	353/51	9/12	199.1	−57.6	232.4	−20.5	41.0	8.1
Bs25	32°30′56″	107°43′40″	31/35	6/10	219.9	−27.7	234.5	−17.8	25.8	13.4
Bs26	32°30′56″	107°43′40″	31/35	4/9	228.9	−25.4	240.0	−11.3	26.3	18.3
Bs27	32°30′41″	107°43′46″	0/55	8/12	23.0	63.5	64.1	21.9	21.6	12.2
Bs34	32°30′16″	107°47′44″	348/54	7/9	180.2	−74.8	240.9	−31.6	93.6	6.3
Bs35	32°30′16″	107°47′44″	357/51	6/12	172.8	−53.3	222.3	−32.8	83.0	7.4
Bs36	32°30′16″	107°47′44″	356/52	9/12	3.3	52.5	44.4	25.5	67.3	6.3
Bs42	32°31′29″	107°47′56″	307/35	7/12	57.7	53.5	50.1	19.8	108.5	5.8
Mean				9	26.6	54.4	53.7	21.6	64.4	6.5

Slat/slong: latitude and longitude of sampling site; N/n: number of samples used to calculate mean/measured; Dg, Ig, Ds, and Is: declination and inclination in geographic and stratigaphic coordinates, respectively; *k*: precision parameter; α_{95}: half angle of cone of 95% confidence about the mean direction after tilt correction. The site mean direction of higher temperature component passed fold test (McFadden, 1990; Enkin, 2003) and reversal test (McFadden and McElhinny, 1990). The parameters of McFadden fold test (1990) are of $\xi_{2(\text{in situ})}$ = 3.794 in geographic coordinates and $\xi_{2(\text{tilt corrected})}$ = 2.477 after tilt correction, while the critical value at 95% is ξ = 3.497. In the fold test of Enkin (2003), the optimal untilting for the Triassic data is WE 116 ± 23% and DC 108 ± 42%. The reversal test of McFadden and McElhinny (1990) is indicated by mean direction of normal direction: Dm/Im = 52.9/22.6, n = 3, k = 68.8; mean direction of reversed direction: Dm/Im = 234.1/−21.1, n = 6, k = 53.1; the angle between these two averages is Gamma = 1.9°.

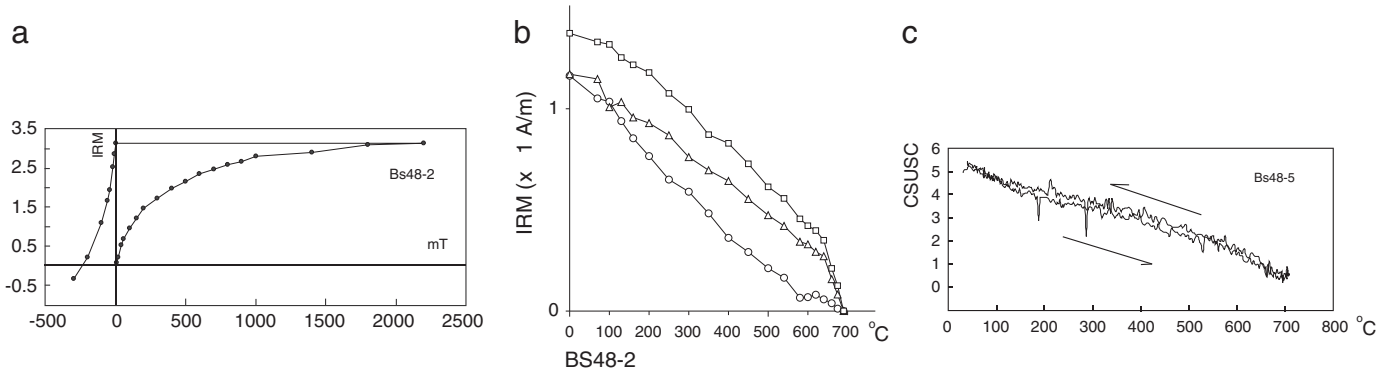


Fig. 5. (a) Isothermal remanent magnetization (IRM) acquisition curves of the Cretaceous samples. (b) Three component IRM thermal demagnetization showing unblocking temperature around 680–690 °C. (c) Temperature dependence of magnetic susceptibility when heating from room temperature to 600 or 700 °C and subsequently cooling curve. The results of thermal-magnetic process indicated that few distinct chemical changes occurred during heating.

and were omitted for further consideration. For most of Cretaceous samples, two components were recognized (Fig. 6). The low temperature components (LTC) were separated below 300 °C–520 °C. Their site-mean direction (before tilt correction: Dg/Ig = 0.7/46.7°, kg = 137.9, α_{95}g = 3.7°; after tilt correction: Ds/Is = 2.2/43.8°, ks = 20.6, α_{95}s = 9.8°) failed the fold test and closed to present geomagnetic field (Fig. 7a, b), which suggested that the LTC represented viscous remanent magnetization. The HTC were separated around 680 °C–690 °C and had a site-mean directions (before tilt correction: Dg/Ig = 25.4/28.1°, kg = 18.4, α_{95}g = 12.3°; after tilt correction: Ds/Is = 28.9/25.7°, ks = 62.7, α_{95}s = 6.6°) (Fig. 7c, d, Table 2) that passed both fold and reversed tests (Enkin, 2003; McFadden, 1990; McFadden and McElhinny, 1990). The maximum precision parameter of the mean HTC direction occurred at 95% unfolding. Thus, we considered that the HTC direction of lower Cretaceous was a primary remanent magnetization.

4. Discussion

Using these primary remanent magnetizations, the corresponding Lower Triassic and Lower Cretaceous poles were calculated (T1: 36.5/208.3°, A_{95} = 5.9°; K1: 57.7/225.8°, A_{95} = 6.0°). These results were then compared with coeval poles from areas around the Sichuan basin (Table 3; Fig. 8). The Lower Triassic and Lower Cretaceous poles obtained from this studies showed distinct clockwise rotations relative to the Sichuan basin (T1, 9.3 ± 5.3°; K1, 8.8 ± 8.3°). Considering another clockwise rotation (26.5 ± 9°) detected by our previous work (Wang and Zhang, 2010) carried out on the early–middle Jurassic strata, the Mesozoic strata in our research area showed a series of clockwise rotations.

Recently, research into the geological structure, petrochemistry and geochronology of the Daba Mountains (S. Dong et al., 2008; Y. Dong

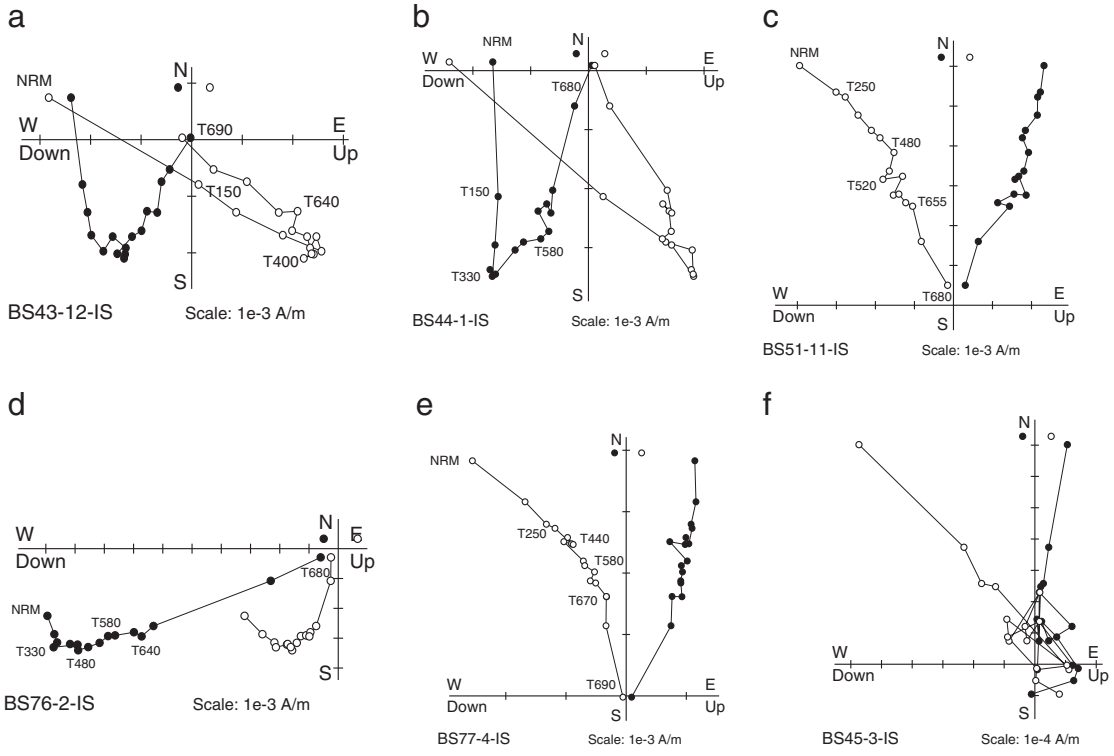


Fig. 6. Representative Zijderveld diagrams of thermal demagnetization of natural remanent magnetism (NRM) from the Cretaceous samples (in in situ coordinates).

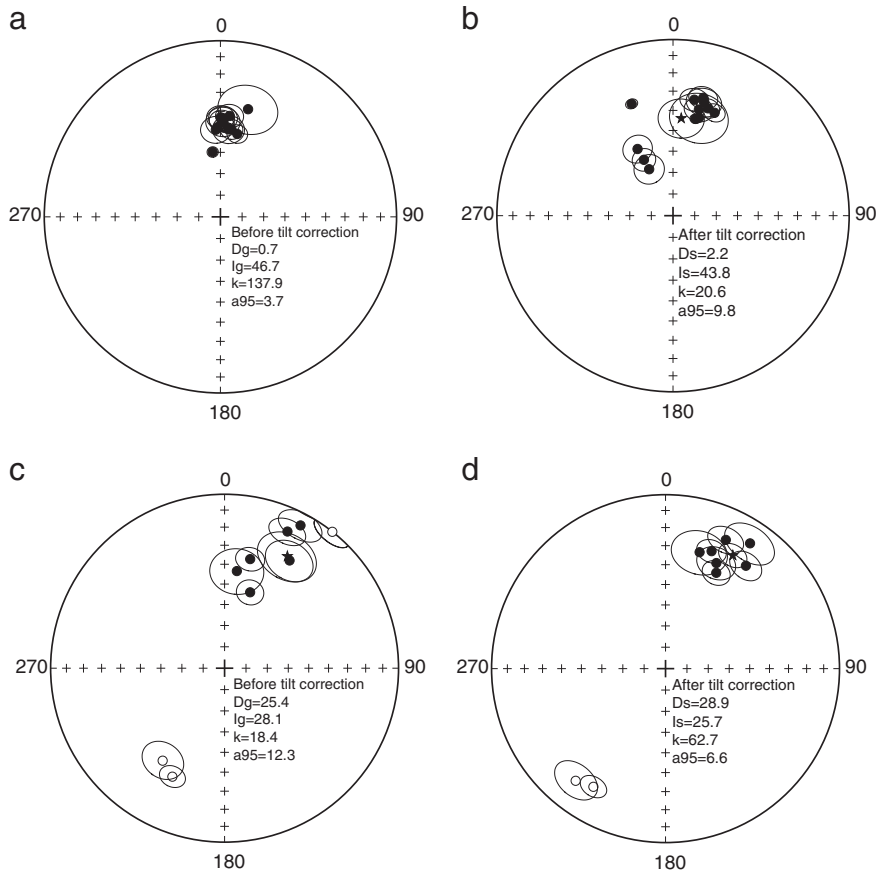


Fig. 7. Equal-area stereographic projections of low temperature components (LTC) (a, b) and high temperature components (HTC); (c, d) site-mean directions from the Cretaceous samples before and after tilt correction.

et al., 2008; He et al., 1997, 1999; Li et al., 2006; Liu et al., 2006; Pei, 2007; Shi et al., 2007; Shen et al., 2007; Wang et al., 2004, 2006; Zhang et al., 2009) has identified a progressive southward-weakening of deformation and youngling fold-related uplift from northeast to southwest (Fig. 9a1–a3). These previous research suggested that the formation of this arc-shaped structural belt was related to detachment and southwestward thrusting process that was obstructed by the rigid Hannan and Huangling–Shennongjia massifs (Guo et al., 1996; Liu et al., 2006; Zhang et al., 2010). The clockwise rotations detected in this study were very probably conducted by this thrusting process. Here, we describe a simple scenario for the deformation in this area (Fig. 9b1–b3), whereby the rigid Hannan massif counteracted thrusting within the northwestern part of the southern Daba Mountains, and contemporaneous deformation along detachment layers to the south resulted in the tectonic rotation of the sampling areas. It is worth noting that the Lower Cretaceous sediments in this area have gentle dips, even though the poles for these sediments provide evidence of rotation. This suggests that rotation occurred before folding during this progressive deformation. In addition, the northeastern part of the Sichuan Basin was split into several smaller areas by the thrust faults, which also involved a dextral strike-slip component (Fig. 9c), as reported by Hu et al. (2009). It's well known that intensive intra-continent subduction occurred in the southern Qinling and Northern Dabashan Mountains during the Late Jurassic and Early Cretaceous. This tectonic process was presumably caused by a combination of the multiple collision that occurred between the Siberia–Mongolia and North China blocks to the

Table 2
Paleomagnetic results of characteristic directions from lower Cretaceous strata.

Site	Slat.	Slong.	Strike	Dip	N/n	Dg	Ig	Ds	Is	*k*	α_{95}
Bs44	32°7′35″	107°20′18″	347	15	13/13	205.7	−31.4	211.4	−21.5	58.2	5.5
Bs45	32°7′35″	107°20′18″	347	15	13/14	13.2	35.8	21.4	28.3	47.9	6.1
Bs46	32°7′35″	107°20′18″	326	17	7/12	7.2	43.1	16.3	30.9	28.2	11.6
Bs47	32°7′35″	107°20′18″	326	17	11/12	18.6	52.0	27.9	37.7	57.8	6.1
Bs48	32°7′25″	107°19′39″	334	20	8/13	213.8	−36.2	218.7	−18.5	37.1	9.2
Bs50	32°7′21″	107°19′12″	329	16	8/9	31.1	28.5	34.0	14.2	30.3	10.2
Bs51	32°3′30″	107°18′43″	99	5	11/13	24.7	14.6	25.1	19.4	41.0	7.2
Bs77	32°14′24″	107°26′18″	130	26	12/12	28.0	7.8	25.7	33.1	28.2	8.3
Bs78	32°14′24″	107°26′18″	130	26	12/12	38.3	−0.1	38.0	25.9	39.0	7.0
Mean					9	25.4	28.1	28.9	25.7	62.7	6.6

Abbreviations are the same as in Table 1. The site mean direction of higher temperature component passed fold test (McFadden, 1990; Enkin, 2003) and reversal test (McFadden and McElhinny, 1990). In the fold test of McFadden (1990), the parameters of McFadden fold test (1990) are of $\xi_{1(\text{in situ})} = 4.236$ in geographic coordinates and $\xi_{1(\text{tilt corrected})} = 3.354$ after tilt correction, while the critical value at 95% is $\xi = 3.497$. In the fold test of Enkin (2003), the Cretaceous optimal untilting is 93 ± 12%, and the Enkin DC untilting is 93 ± 36%. The reversal test of McFadden and McElhinny (1990), mean direction of normal direction: Dm/Im = 27.1/27.2, n = 7, k = 61.0; mean direction of reversed direction: Dm/Im = 215.1/−20.0, n = 2, k = 234.4; the angle between these two averages is 10.3°.

Table 3
Available Early Triassic and Early Cretaceous poles around the Sichuan basin.

Age	Sampling locality		Direction			Pole		A_{95}	Reference
	°N	°E	Dec.	Inc.	α_{95}	λ (°N)	Φ (° E)		
K1	26.0	117.2	23.6	33.7	7.3	66.9	221.4	5.4	Zhai et al. (1992)
K1	26.8	102.5	22.1	37.1	3.8	69.0	204.6	4.3	Huang and Opdyke (1992)
K1	30.0	102.9	13.3	36.4	4.0	74.5	229.0	3.2	Enkin et al. (1991)
K1	32.1	107.4	28.9	25.7	6.6	57.7	225.8	6.0	This study
Mean (N = 3) 70.4/217.5, α_{95} = 8.7									
T1	28.0	105.5	42.1	10.2	6.8	44.1	217.2	4.9	Yang and Besse (2001)
T1	29.9	106.3	49.4	14.9		38.7	211.8	5.6	Steiner et al. (1989)
T1	29.6	103.4	36.8	14.4	3.5	48.8	218.7	3.1	Ma and Zhang (1989)
T1	29.3	107.7	42.8	14.7	3.9	44.3	217.4	3.5	Heller et al. (1995)
T1	32.1	106.2	45.1	18.0	5.9	42.5	214.6	8.5	Bai et al. (1997)
T1	32.5	107.7	53.7	21.6	6.5	36.5	208.3	5.9	This study
Mean (N = 5) 43.7/215.8, α_{95} = 3.9									

north, between the North China and Lhasa–Qiangtang–Indochina blocks to the west, and the subduction of oceanic crust of the Pacific Plate to the east (Dong et al., 2011; Ratschbacher et al., 2003). Data of ESR (Electron Spin Resonance) on quartz of the conjugate joint and fission track (Liu et al., 2006; Shen et al., 2007) indicated that the majority of folding within the South-Daba Mountains mainly occurred during the Early Cretaceous. This suggested that the tectonic rotation detected in this research might have resulted from long-term tectonic events that peaked in the Early Cretaceous, sometime after the deposition of the Cangxi Formation. However, it was not possible to determine the timing of the end of this long-term process, as the study area does not contain exposed Late Cretaceous and younger sediments.

The amalgamation between the NCB and SCB is thought to have ceased around the Early–Middle Jurassic (Yang et al., 1992; Yang and Besse, 2001), although subsequent intracontinental tectonic process occurred within the NCB and SCB until the Early Cretaceous. This suggests that the rotations detected by our results may have been associated with the initial stages of this tectonic developing process, and further research in different areas of the Daba Mountains is needed to test the oroclinal–rotation hypothesis.

Paleomagnetic data from Triassic units indicated that the central Yangtze Fold Belt also underwent a similar clockwise rotation relative to the Sichuan basin (Huang and Opdyke, 1996, 1997; Steiner et al., 1989; Su et al., 2005; Tan et al., 2007). Tan et al. (2007) argued that this rotation was probably related to the collision between the NCB and SCB. However, the timing of this clockwise rotation is still unknown, and the limited number of Jurassic and Cretaceous paleopoles for this area means that a detailed rotation model cannot be constructed. However, stratigraphy and thermochronology can be used to provide alternative evidence for the rotation of this area. Liu et al. (2005) suggested that a continuous belt of basins, termed the northern Yangtze Basin developed along the southern front of the Qinling–Dabieshan during the Early–Middle Jurassic with the northeastern Sichuan, Zigui and Dangyang Basins also filled by the same depositional system. Southward-directed thrusting of the northern Yangtze fold thrust belt (NYFB) and northwestward-directed thrusting of the Jiangnan fold thrust belt (JNFB) caused Later Jurassic to the Early Cretaceous uplift of the Zigui Basin uplifted, resulting in the migration of the depocenter to the northern Sichuan Basin. In addition, apatite fission-track dating indicates that the Tongbai–Dabie region, Huangling and the Daba Mountains areas underwent rapid uplifting and cooling event during the Later Jurassic to Early Cretaceous (Hu et al., 2006; Shen et al., 2007, 2009). This was followed by a period of relative thermal stability before accelerated uplifting in the Cenozoic. This suggests that the tectonic rotations within both the Middle Yangtze Fold Belt and the Daba Mountains were the result of similar processes, with both being affected by late Mesozoic intracontinental deformation associated with long-term tectonism rather than a single tectonic event.

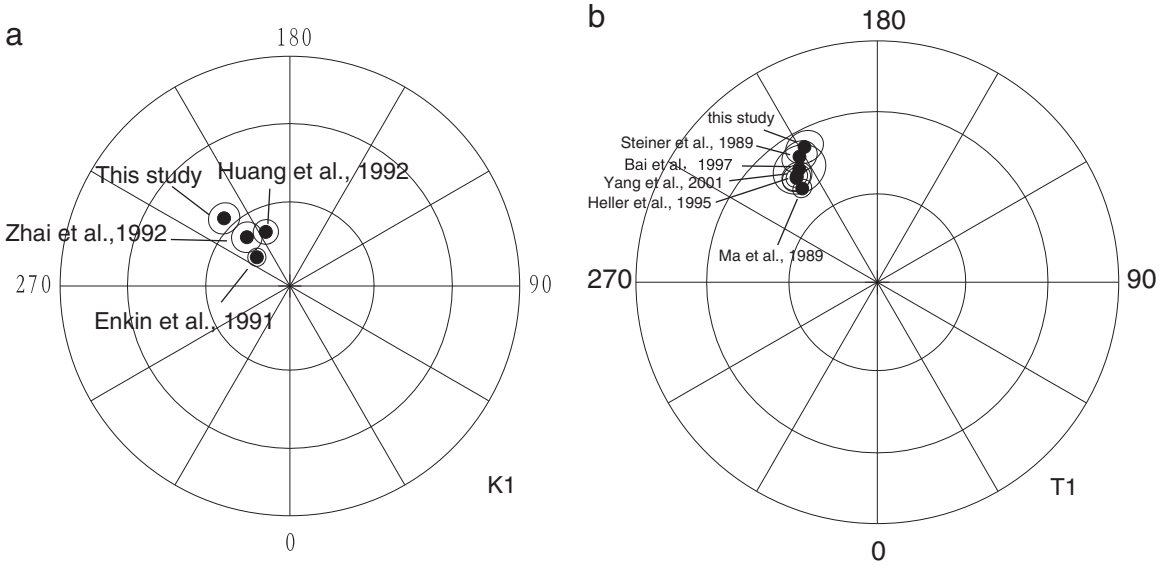


Fig. 8. Equal-area projections of the early Cretaceous (a) and Triassic (b) paleopoles with circles of 95% confidence obtained from the Sichuan basin.

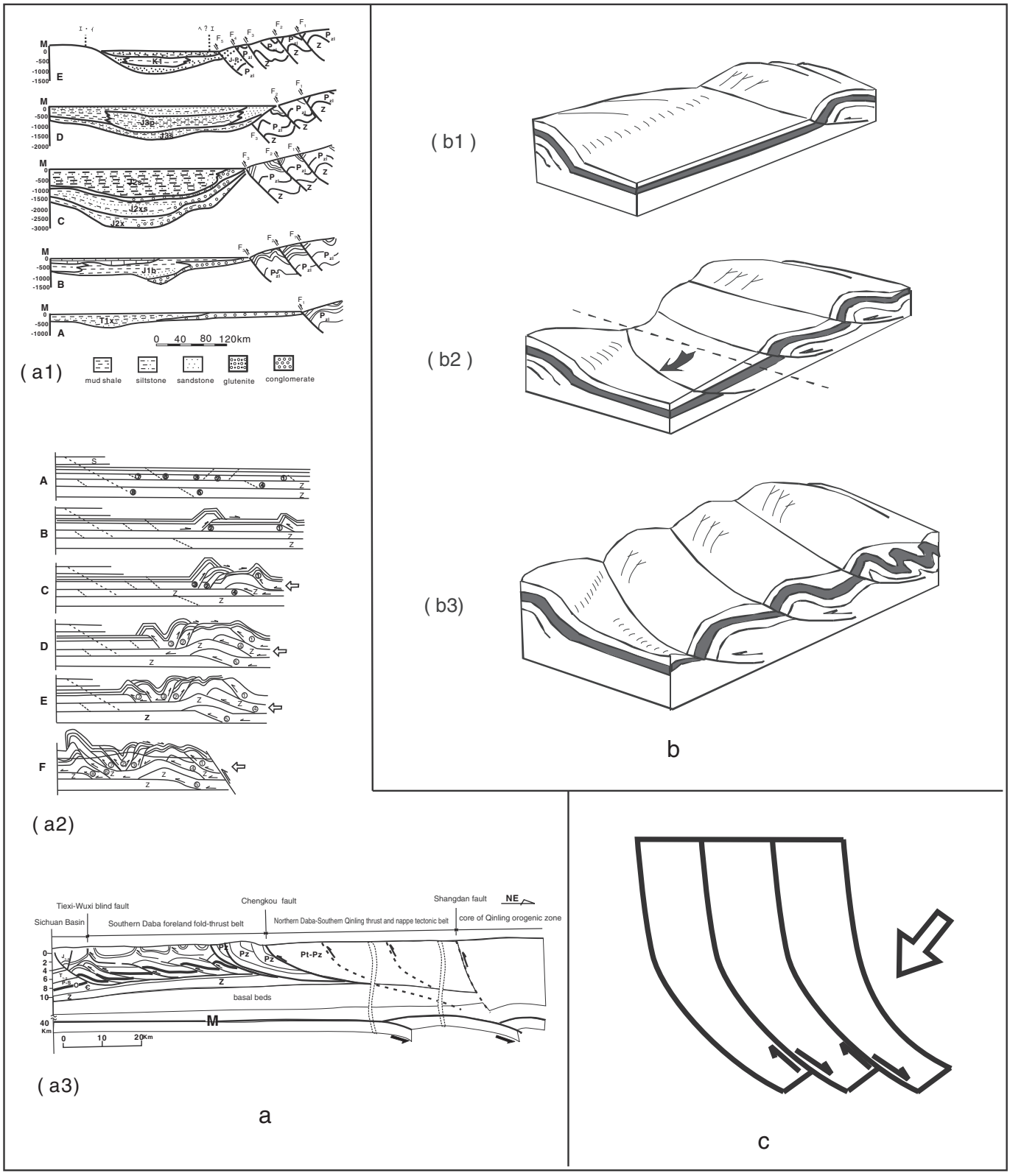


Fig. 9. (a) Tectonic evolution models proposed for the Southern Dabashan foreland thrust belt ((a1) from Yue, 1988, (a2) from He et al., 1997, (a3) from Li et al., 2006). (b) Simple sketch map explains the tectonic rotation near the northeastern Sichuan Basin and the southern Daba Mountain. (c) Schematic diagram showing the formation of thrusting fault with dextral shear component in the Southern Daba Mountain and adjacent area.

5. Conclusion

Two new Early Triassic and Early Cretaceous paleomagnetic poles were obtained from the northeastern Sichuan basin and the southern Daba Mountains. These data indicate that this area underwent a series of clockwise tectonic rotations relative to reference poles in the Sichuan Basin. The similarities between the Daba Mountains and the central Yangtze Fold Belt suggest that the tectonic rotations of both fold belts

were caused by post-collision deformation of the amalgamated NCB and SCB, as part of long-range tectonism rather than a single tectonic event.

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